Chapter 2

Capacity Utilization in the Center and the Periphery: Old Debates and New Estimates

Dissertation Chapter — Canonical Scaffold

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Abstract

[DRAFT PROSE: Abstract: 150–200 words. State: (1) the structural deficiency of existing CU estimates and why it matters for crisis theory; (2) the RRR/CVAR identification strategy and what it recovers; (3) the behavioral recapitalization law and its nested closures; (4) the Tavares peripheral twist as the chapter's distinctive contribution; (5) the three downstream objects inherited by Ch3.]

1 Introduction

[DRAFT PROSE: §2.1 Introduction, approx. 3–4 pages. Lead with: Ch1 showed existing CU estimates impose balanced growth by construction ($\theta = 1$), suppressing regime-dependent content. Ch2 asks: *what does CU do in the economy?* Following Okishio (1961): CU path is a crisis trigger. Then: (i) identify CU as latent confinement object of the long-run reproduction structure via RRR/CVAR; (ii) estimate behavioral recapitalization law conditional on recovered objects. Run same machine on US and Chile. The BoP constraint enters the recapitalization process differentially in the periphery: you have the surplus but cannot buy the machines (?). Three downstream objects: $\hat{\mu}_t$, $\hat{B}_t/\hat{\theta}_t$, behavioral accumulation law with tested closures. Roadmap paragraph at end.]

The remainder of the chapter is organized as follows. Section 2 establishes the crisis-theoretic motivation. Section 3 develops the accounting scaffold. Section 4 introduces the distributive micro-foundation. Section 5 bridges the Ch1 focal data pair to Ch2's state vector. Section 6 estimates the structural identification (Stage A). Section 7 derives the accounting law of capital accumulation. Section 8 estimates the behavioral recapitalization law (Stage B). Section 9 formalizes the peripheral twist. Section 10 synthesizes the center–periphery comparison. Section 11 concludes.

2 Crisis Theory: Stagnation Tendencies, Cumulative Disequilibrium, and the Role of Capacity Utilization

[DRAFT PROSE: §2.2 intro paragraph: Establish that CU is the mechanism through which accumulation dynamics trigger crisis. Ch1 established *measurement* failure; this section establishes *why it matters*.]

2.1 Stagnation and Crisis: A Typology

[DRAFT PROSE: §2.2.1: 1–2 pages. Present the Vidal (2014, 2019) / Basu (2019) typology. Distinguish partial crisis (interruption resolvable by partial institutional reconfiguration of Λ) from structural crisis (deep interruption requiring drastic restructuring). Include Table 1.]

Table 1: Stagnation Tendencies: A Typology

	Excess of Surplus Value	Deficit of Surplus Value
Demand side	Under-consumption / realization crisis	Profit squeeze (rising labor strength)
Financial side	Financial fragility	Falling (normal) rate of profit

Notes: Following **??**. Each cell denotes a qualitatively distinct stagnation tendency. Partial crisis $\equiv$ interruption resolvable by partial reconfiguration of the institutional space Λ. Structural crisis $\equiv$ deep interruption requiring drastic restructuring.

2.2 Okishio's Crisis Trigger and Cumulative Disequilibrium

[DRAFT PROSE: §2.2.2: 1.5 pages. Okishio (1961): decentralized decisions produce secular cumulative disequilibrium in CU. The path of CU (not a static level) is the crisis trigger. Connect to Weisskopf's decomposition $r = \mu b \pi$: profitability is demand-led, so the CU path directly mediates the link between accumulation dynamics and crisis episodes. Note: μ here corresponds to our μ_t — utilization rate, not the mathematical constant.]

The **?** (**?**) decomposition of the profit rate furnishes a compact expression for the demand-mediation channel:

$$r_t = \mu_t \cdot B_t \cdot \pi_t, \tag{1}$$

where $r_t \equiv \Pi_t/K_t$ is the profit rate, $\mu_t \equiv Y_t/Y_t^p$ is capacity utilization, $B_t \equiv Y_t^p/K_t$ is capital productivity at normal capacity, and $\pi_t \equiv \Pi_t/Y_t$ is the profit share. **[DRAFT PROSE: Continue: the crisis trigger is the path of μ_t, not a static level. Foreshadow**

the identification problem: to read (1) as an empirical relationship, one needs a structurally identified μ_t.]

2.3 From Profitability to Accumulation: The Missing Link

[DRAFT PROSE: §2.2.3: 1 page. The literature has studied the tendency of the profit rate to fall (??) but the causal link from profitability to crisis runs through accumulation demand (???). This chapter formalizes that link through the behavioral recapitalization law. The mechanism differs structurally between center and periphery (Tavares 1985 — foreshadow §2.9). Transition sentence to §2.3.]

3 Structural Primitives: Production, Capacity, and Unbalanced Growth

[DRAFT PROSE: §2.3 intro paragraph: This section establishes the accounting scaffold. All steps are pure accounting — no behavioral equation is imposed until §2.7. Emphasize the distinction between the accounting identity and the behavioral closure (a load-bearing architectural separation).]

We adopt a Leontief production structure in which both capital and labor bind at full capacity utilization:

$$Y_t = \min(B_t K_t,\ A_t L_t)\,, \tag{2}$$

with both constraints binding at $\mu_t = 1$. Output decomposes into productive capacity and utilization:

$$Y_t = \mu_t \cdot Y_t^p, \tag{3}$$

$$y_t = y_t^p + \ln \mu_t, \tag{4}$$

where lowercase denotes natural logarithms of the corresponding uppercase level, $y_t \equiv \ln Y_t$, $k_t \equiv \ln K_t$. Capital productivity at normal capacity is

$$B_t \equiv \frac{Y_t^p}{K_t}, \qquad \ln B_t = y_t^p - k_t. \tag{5}$$

The long-run transformation of capital stock into productive capacity is governed by the *unbalanced growth closure*:

$$y_t^p = \theta(\Lambda) \cdot k_t, \tag{6}$$

where $\theta(\Lambda)$ is the capacity transformation elasticity — a regime parameter, pinned by the institutional space Λ and not derivable from within-period optimization. Balanced growth ($\theta(\Lambda) = 1$) is the knife-edge special case; the general case is $\theta(\Lambda) \neq 1$.

Proposition 3.1 (Confinement Identity)**.** *Given the decomposition* (4) *and the unbalanced growth closure* (6)*,*

$$\ln \mu_t = y_t - \theta(\Lambda)\, k_t. \tag{7}$$

Capacity utilization is identified as the deviation of realized output from the productive-capacity manifold.

[DRAFT PROSE: §2.3 remainder: explain the economic content of Proposition 3.1. Emphasize: (i) this is an accounting identity, not a behavioral claim; (ii) $\theta \neq 1$ implies persistent structural disequilibrium in the capital–output ratio; (iii) the Harrodian benchmark corresponds to $\theta = 1$ (the knife-edge), which existing CU measures impose by construction. Transition to §2.4: to estimate (7), we need an empirically tractable specification of θ.]

4 Distribution-Conditioned Capacity: The Interaction Term

[DRAFT PROSE: §2.4 intro paragraph: Introduce the distributive micro-foundation. The transformation elasticity is not fixed — it varies with the profit share through the mechanization incentive (FMT).]

Let the exploitation rate e_t map into the profit share as

$$\pi_t = \frac{e_t}{1 + e_t}, \tag{8}$$

and introduce the interaction term $z_t \equiv \pi_t \cdot k_t$ as the empirical proxy for $\theta(e|\Lambda) \cdot k_t$ in the cointegrating space. The long-run capacity relation then takes the estimable form:

$$y_t = c + \beta_1 k_t + \beta_2(\pi_t k_t) + \xi_t, \qquad \xi_t \equiv \ln \mu_t, \tag{9}$$

so that the endogenous long-run transformation elasticity is

$$\theta_t = \beta_1 + \beta_2 \, \pi_t. \tag{10}$$

Proposition 4.1 (Rank-1 Identification)**.** *If $\xi_t = y_t - c - \beta_1 k_t - \beta_2(\pi_t k_t)$ is stationary, then*

$$\widehat{\ln \mu_t} = y_t - \hat{c} - \hat{\beta}_1 k_t - \hat{\beta}_2(\pi_t k_t) \tag{11}$$

is the recovered latent utilization object. This is the dissertation's primary identification problem.

[DRAFT PROSE: §2.4 remainder: three points. (1) θ is upstream: β_1 and β_2 are structural parameters of the cointegrating relation, not behavioral choices. The linearization (10) is the empirical proxy for the general $\theta(e|\Lambda)$; the quadratic

approximation $\theta_0 + \theta_1 e + \theta_2 e^2$ from `theory_estimation_mapping.md` is the theoretical object it approximates. (2) The interaction term $z_t = \pi_t k_t$ rotates the cointegrating space: it is a constructed object needed to represent $\theta(e) \cdot k$ in the VECM, not a primitive state variable. (3) IICS as a falsifiable hypothesis: the sign and magnitude of β_2 determine whether higher exploitation raises or depresses the capacity-transformation elasticity. Transition to §2.5.]

5 From Chapter 1 to Chapter 2: The Focal Data Pair and the State Vector

[DRAFT PROSE: §2.5 intro: 1 paragraph bridging Ch1 to Ch2. Ch1 established VAcorp/KGCcorp as the empirically validated CU proxy pair for the US; ran ARDL bounds tests and Shaikh-style replication (S0–S2). This section maps the Ch1 focal pair into Ch2's cointegrating state vector and extends the logic to Chile.]

5.1 US: From Ch1's ARDL to Ch2's Cointegrating Relation

[DRAFT PROSE: §2.5.1: 1.5 pages. Ch1's bivariate ARDL on (y_t, k_t) with VAcorp/KGCcorp recovered $\hat{\theta}^{S1}$ and diagnosed rank-1 confinement (S2). Ch2 augments with $z_t = \pi_t k_t$, allowing θ_t to vary with distribution. State vector: $\mathbf{X}_t^{US} = (y_t,\, k_t,\, \pi_t k_t)'$. Data: BEA/FRED (y, k) from Ch1; NIPA Table 1.12 $(e \to \pi)$. Sample: 1947–2019. Positioning: contrast with CBO, Adams–Coe, multivariate filter methods that impose $\theta = 1$.]

The state vector for the US system is

$$\mathbf{X}_t^{US} = \begin{pmatrix} y_t \\ k_t \\ \pi_t k_t \end{pmatrix}, \tag{12}$$

estimated on annual data from 1947 to 2019, with output and capital stock inherited from the Chapter 1 BEA/FRED pipeline, and the profit share constructed from NIPA Table 1.12.

5.2 Chile: Constructing the Parallel

[DRAFT PROSE: §2.5.2: 2 pages. Parallel data construction: y_t from Central Bank / PWT real GDP; k_t from PWT `rnna` or Hofman (2000)/Lüders et al. (2016)/ Pérez-Eyzaguirre (2019) spliced series; $e_t \to \pi_t$ from national accounts spliced with Alarco Tosoni (2014) and Astorga (2023) wage share series. Sample: 1960–2019. Positioning against Chilean CU literature: BCCh structural balance framework, Magud–Medina, Fuentes–Gredig–Larraín; emphasize that Ch2's identification permits $\theta \neq 1$ and recovers the distribution-conditioned slope — an *alternative* to the governance variable embedded in the structural fiscal rule.]

The state vector for the Chilean system mirrors (12):

$$\mathbf{X}_t^{CL} = \begin{pmatrix} y_t^{CL} \\ k_t^{CL} \\ \pi_t^{CL} k_t^{CL} \end{pmatrix}. \tag{13}$$

5.3 The Parallel Construction as Method

[DRAFT PROSE: §2.5.3: 0.5 page. Emphasize: same $\mathbf{X}_t$, same Johansen procedure, same rank-1 identification for both countries. The structural comparison

in §2.10 is meaningful *because* the measurement apparatus is identical. The key asymmetries (US budget-baseline embedding vs. Chilean structural-balance embedding) are *findings* of the comparison, not artefacts of measurement. Transition to §2.6.]

6 Stage A: Structural Identification (RRR/CVAR)

[DRAFT PROSE: §2.6 intro: 1 paragraph. State the estimating logic: reduced-rank regression on $\mathbf{X}_t = (y_t, k_t, z_t)'$ where $z_t = \pi_t k_t$. Methodology: Johansen (1988) trace and max-eigenvalue tests; restricted constant specification; Gregory-Hansen endogenous break detection.]

6.1 Estimation

[DRAFT PROSE: §2.6.1: 1 page. Standard Johansen setup. Trace test $H_0 : r \leq j$ vs. $H_1 : r > j$ for $j = 0, 1, 2$. Expected rank: $r = 1$ (rank-1 confinement hypothesis). Finite-sample critical values (Johansen–Juselius 1990). Restricted constant specification: drift absorbed into the cointegrating space, not the common trends. Break detection: Gregory–Hansen (1996) for structural break at unknown date — relevant given the 1973/1982 Chilean regime shifts.]

6.2 Results: US Baseline

[DRAFT PROSE: §2.6.2: 2–3 pages (to be filled after estimation). Report: $\hat{\beta}_1$, $\hat{\beta}_2 \Rightarrow \hat{\theta}_t^{US}$. Recovered $\widehat{\ln \mu_t^{US}}$. Regime classification: does $\hat{\theta}^{US}$ cross unity? When? Loading coefficients and speed of adjustment. Diagnostics: CUSUM, recursive eigenvalues.]

6.3 Results: Chile

[DRAFT PROSE: §2.6.3: 2–3 pages (to be filled after estimation). Same structure as §2.6.2. First structural comparison: $\hat{\theta}^{US}$ vs. $\hat{\theta}^{CL}$ — the mechanization gap in θ-units. Discuss regime breaks: pre-/post-1973, pre-/post-1982.]

6.4 Derived Structural Objects

Once the cointegrating parameters are recovered, the following structural objects are constructed by definition:

Definition 6.1 (Recovered Productive Capacity)**.**

$$\hat{y}_t^p = y_t - \widehat{\ln \mu_t}. \tag{14}$$

Definition 6.2 (Recovered Capital Productivity at Normal Capacity)**.**

$$\widehat{\ln B_t} = \hat{y}_t^p - k_t, \qquad \hat{B}_t = \exp(\hat{y}_t^p - k_t)\,. \tag{15}$$

These are the Stage B inputs: $\widehat{\mu}_t$ and $\hat{B}_t$ are passed into the behavioral accumulation law.

[DRAFT PROSE: §2.6.4 narrative: emphasize the two-stage architecture. Stage A recovers $\hat{\mu}$ and $\hat{B}$ cleanly only after the identification problem is solved. An ARDL investment equation cannot perform this identification — it can only describe how accumulation responds to utilization once utilization has been structurally recovered.]

7 The Accounting Law of Capital Accumulation

[DRAFT PROSE: §2.7 intro: 1 paragraph. This section bridges Stage A (identification) to Stage B (behavioral estimation). The accounting law is derived from pure identity manipulation — no behavioral content is imposed. The behavioral margin is whether the recapitalization rate β_t is constant.]

Define the gross accumulation rate as

$$\iota_t \equiv \frac{I_t}{K_t}. \tag{16}$$

The accounting decomposition of accumulation is:

$$\iota_t = \phi_t \cdot \mu_t \cdot B_t, \qquad \phi_t \equiv \frac{I_t}{Y_t}, \tag{17}$$

and since $\phi_t \cdot \mu_t = (I_t/Y_t)(Y_t/Y_t^p) = I_t/Y_t^p \equiv \phi_t^p$, we have the equivalent form

$$\iota_t = \phi_t^p \cdot B_t, \qquad \phi_t^p \equiv \frac{I_t}{Y_t^p}, \tag{18}$$

where utilization drops out at this level of abstraction. Introducing the recapitalization rate $\beta_t \equiv I_t/\Pi_t$, and using $\pi_t \equiv \Pi_t/Y_t$ and $r_t \equiv \Pi_t/K_t = \pi_t \mu_t B_t$:

$$\boxed{\iota_t = \beta_t \cdot \pi_t \cdot \mu_t \cdot B_t = \beta_t \cdot r_t.} \tag{19}$$

Equation (19) is an identity. There are two recapitalization lenses:

- $\beta_t = I_t/\Pi_t$: funding of investment out of profits;
- $\phi_t^p = I_t/Y_t^p$: claim of investment on productive-capacity output.

They are linked accounting representations of the same accumulation law.

[DRAFT PROSE: §2.7 remainder: emphasize the dual-lens interpretation.

The behavioral content enters through the question of whether β_t is constant. If it is, the accounting decomposition holds mechanically and accumulation is fully determined by the structural objects. If not, the deviation $\eta_j = b_j - 1$ carries the behavioral information. Note the Weisskopf decomposition (1) as a special case. Transition: §2.8 makes β_t the object of behavioral estimation.]

8 Stage B: Behavioral Accumulation Law (ARDL)

[DRAFT PROSE: §2.8 intro: 1 paragraph. Stage B treats the recovered structural objects $(\hat{\mu}_t, \hat{B}_t)$ as given and estimates how capitalists behaviorally determine the recapitalization rate. This is a conditional behavioral module nested inside the structural-identification architecture.]

8.1 Specification

If recapitalization responds behaviorally to distribution, demand, and structural conditions, then

$$\ln \beta_t = \eta_0 + \eta_1 \ln \pi_t + \eta_2 \ln \mu_t + \eta_3 \ln B_t, \tag{20}$$

which, substituted into the log of (19), yields the long-run behavioral accumulation law:

$$\boxed{\ln \iota_t = c + b_1 \ln \pi_t + b_2 \ln \mu_t + b_3 \ln B_t + \varepsilon_t} \tag{21}$$

with $b_j = 1 + \eta_j$.

Proposition 8.1 (Dual Interpretation of Coefficients)**.** *Each coefficient in* (21) *has both an accounting and a behavioral reading:*

8.2 Nested Restriction Menu

The behavioral coefficients generate a hierarchy of accumulation closures:

Table 2: Dual Interpretation of Accumulation Law Coefficients

Coefficient	Accounting reading	Behavioral reading
b_1	Profit-share elasticity of accumulation	$\eta_1 = b_1 - 1$: recapitalization response to di
b_2	Utilization elasticity of accumulation	$\eta_2 = b_2 - 1$: recapitalization response to de
b_3	Capital-productivity elasticity of accumulation	$\eta_3 = b_3 - 1$: recapitalization response to st

Notes: Each $b_j = 1 + \eta_j$. The deviation from unity is the behavioral content. Estimation uses ARDL with Pesaran–Shin–Smith (2001) bounds test.

Table 3: Nested Restriction Menu

Restriction	Null	Model name
Cambridge closure	$b_1 = b_2 = b_3 = 1$	β_t constant
Bhaduri–Marglin type	$b_3 = 1$; b_1, b_2 free	Blind to capital productivity
Full structural model	All b_j free	All channels active

8.3 Results: US

[DRAFT PROSE: §2.8.3: 2–3 pages (to be filled after estimation). ARDL estimation with PSS bounds test. Report $\hat{b}_1, \hat{b}_2, \hat{b}_3$ and associated Wald tests. Which closure is supported? Interpret $\eta_j = b_j - 1$ economically.]

8.4 Results: Chile

[DRAFT PROSE: §2.8.4: 2–3 pages (to be filled after estimation). Same procedure. Key question: does b_3 behave differently from the US? Does the capital-productivity channel operate freely or is it structurally constrained? Transition: §2.9 provides the mechanism.]

9 The Peripheral Twist: BoP Constraint and the Import Content of Recapitalization

[DRAFT PROSE: §2.9 intro: 1 paragraph. This is the chapter's distinctive theoretical contribution. The Tavares (1985, 2000) channel formalized: the binding constraint on peripheral accumulation is not on output directly but on the ability to convert surplus into re-investment.]

9.1 The Tavares Channel

[DRAFT PROSE: §2.9.1: 2 pages. Mechanism: (1) peripheral accumulation requires imported machinery — capital goods sector structurally incomplete (Tavares's "hard phase" of ISI, Fajnzylber's "truncated industrialization"); (2) the recapitalization rate β_t is structurally conditioned by forex availability: profits exist but recapitalization collapses when imported capital goods are unaffordable; (3) sudden stops in β_t — debt-cycle reversals, terms-of-trade collapses, Volcker shock — express financial dependency as a constraint on the technical composition of accumulated capital. Reference: ?, ?.]

9.2 Formalization

[DRAFT PROSE: §2.9.2: 1.5 pages. The recapitalization response $\eta_3 = b_3 - 1$ is not a free behavioral parameter in the periphery — it is structurally constrained by the import content of investment and the availability of foreign exchange. Candidate specifications: (i) augment Chilean ARDL with forex availability proxy (terms of trade, real exchange rate, current account balance); (ii) test whether b_3^{CL} is stable across BoP regimes or breaks at sudden stops; (iii) compare b_3^{US} (unconstrained) vs. b_3^{CL} (BoP-mediated).]

Let $\tilde{\phi}_t^{ME}$ denote the import content of investment — the share of machinery and equip-

ment in gross capital formation that is sourced from abroad. The peripheral behavioral law gains a forex-mediation channel:

$$\eta_3^{CL} = \eta_3^{CL}\left(\tilde{\phi}_t^{ME}, \mathrm{BoP}_t\right), \tag{22}$$

where BoP_t is a measure of external constraint intensity (terms of trade, real exchange rate, or current account balance).

9.3 The Technical Composition Channel

[DRAFT PROSE: §2.9.3: 1 page. When $\theta < 1$ (over-mechanization), the economy needs *more* imported capital goods per unit of capacity created, but its capacity to earn forex is *lower* because the accumulated capital is less productive. The peripheral trap: over-mechanization raises the import bill for investment while degrading the export capacity that finances it. Link to Prebisch–Singer: the secular deterioration of the terms of trade is not just a price phenomenon — it is a structural constraint on the technical composition of peripheral capital accumulation. Approved punchline: *"demand conditions intervene on the political economy of the ceiling, i.e. the class struggle over distribution, utilization, and the institutional forms that govern whether accumulation can proceed."*]

10 Structural Comparison: Center vs. Periphery

[DRAFT PROSE: §2.10 intro: 1 paragraph. Synthesize the results across both countries. The parallel construction (§2.5.3) ensures the comparison is structural, not an artefact of measurement.]

[DRAFT PROSE: §2.10 narrative: explain each row of Table 4. Punchline: financial dependency shows up empirically as a structurally constrained recapitalization response. The Tavares channel provides the mechanism; Table 4 provides

Table 4: Center–Periphery Structural Comparison

Diagnostic	US	Chile	Reading
$\hat{\theta}$	—	—	Mechanization gap
Regime breaks	—	—	When does θ shift?
b_1 (distribution)	—	—	Profit-share response
b_2 (demand)	—	—	Utilization response
b_3 (structural)	—	—	Capital-productivity response
b_3 stability	—	—	Does b_3^{CL} break at BoP crises?
$\widehat{\mu}_t$ volatility	—	—	Utilization instability
$\hat{B}_t$ trend	—	—	Capital productivity trajectory

Notes: Cells filled after Stage A and Stage B estimation. b_3 is the key comparison: in the US, the recapitalization function responds to the capital-productivity channel relatively freely; in Chile, b_3^{CL} is mediated by the BoP constraint (§2.9).

the evidence.]

11 Conclusions

[DRAFT PROSE: §2.11: approx. 3 pages. Three deliverables: (1) CU series: structurally identified $\hat{\mu}_t$ for both countries, available as inputs to Ch3. (2) Regime classification: $\hat{\theta}^{US}$ vs. $\hat{\theta}^{CL}$ — the empirical realization of the core-periphery mechanization gap. (3) Financial dependency formalized: the Tavares channel — inability to convert surplus into re-investment due to lack of foreign currency — is not a narrative but an estimable structural constraint on the recapitalization function. Forward references: Ch3 inherits $\hat{\mu}_t$, $\hat{B}_t$, $\hat{\theta}_t$ as given inputs for profitability analysis and investment function hypothesis testing. The Weisskopf decomposition and Okishio crisis trigger become Ch3 applications of the Stage A recovered objects.]

References